

Final Project Report

0. Project title:

Short project description - “tag line”

Team member’s names:

Date submitted:

Your selected area of emphasis: Appropriate Framing · Clear Values & Tradeoffs · Creative Alternatives · Relevant & Reliable Information · Sound Reasoning · Commitment to Action

Each team submits one report. Each member should upload an identical copy in your canvas account. Reports that differ among team members will be rejected.

Your final report and your final slide presentation contain the same content, following this template. You can use an AI LLM to convert your final report into a slide presentation. Give it hints by adding comments such as (:slide 10”) to preface each section. See how well it does! Presentations each have a 10 minute time limit, so about 10 slides is the right length.

Replace italicised prompts in this document with your own content; delete the prompts when you submit. Aim for a complete, readable, self-contained account — length is up to you.

1. Your team.

Say a few words about who you are, your background and your interests. You’ve done a lot of work – brag a little!

2. One Page Executive summary: The problem and area of emphasis

Make the case for your analysis in one page.

- Say who the decision maker is
- Summarize the decision and why it matters,
- Explain your model by including your CDN diagram, - perhaps simplified
- Reveal your findings
- Make your recommendation to the decision maker

3. Appropriate Framing — Solving the Right Problem

This section corresponds to the Appropriate Framing area of Decision Quality. It must establish the context of the decision, its scope and its boundaries.

- - — Primary decision statement

State the decision in one or two sentences. A decision must have a decision maker, a set of alternatives, and a time at which it is made. (It is not a question to study or a problem to investigate.)

- - — *Decision maker and perspective*

Who owns this decision and its consequences? If you are acting as decision analyst, describe the decision maker explicitly. State what the decision maker's perspective or span of responsibility is.

- - — *Frame: inside vs. outside scope*

List what is inside the frame (variables, costs, consequences, stakeholders) and what you are treating as outside it. Justify any reframing you did since the proposal.

- - — *Purpose and values served*

What is the decision maker ultimately trying to achieve? One paragraph.

4. Clear Values and Tradeoffs

This section consolidates Task 4 of HW5 (value function and utility elicitation) and the corresponding section of the project draft. Define the value function before attempting any utility assessment.

- - — *Value measure and units*

Name the prospect that captures what the decision maker cares about. If a fungible quantity, e.g. dollars, QALYs, kWh, time saved, an interval scale, explain its relation to the problem. If multiple attributes are present, state explicitly how they combine into one value.

- - — *Value function*

For a financial project: write out the dollar formula — revenue, cost, and discount model — that produces one number for every terminal node. For a non-financial project: define the constructed scale, its degrees, and how outcomes map to it.

- - — *Ethical concerns and tradeoffs*

Identify any values in tension (e.g., cost vs. safety, individual vs. collective). Note any ethical constraint that bounds the alternatives. Even if tradeoffs are straightforward, state that explicitly.

- *INCLUDE IF APPLICABLE* — *Utility elicitation — best and worst prospects*

Required when value is subjective, multi-attribute, or when risk attitude materially affects the choice. Identify the best prospect (utility = 1) and worst prospect (utility = 0). If the value function is purely financial and well-defined, you may state that subjective elicitation is deferred, and explain why.

- — *intermediate prospects*

For at least two intermediate prospects, use the Equivalence Rule with the probability wheel: find the preference probability p that makes the decision maker indifferent between the prospect for certain and a deal [best with p , worst with $1-p$]. Report each elicited utility and confirm the utilities respect the preference ordering.

- *INCLUDE IF APPLICABLE* — *Certain equivalent and risk premium*

For a risky monetary project, calculate a certain equivalents. Compute risk premium = Expected Value – Certain Equivalent. State whether the decision maker is risk averse, risk neutral, or risk seeking, and describe how the utility function bends (e.g., convex upward for risk aversion). If value involves time, state your discounting approach.

5. Creative Alternatives

Enumerate the alternatives your decision maker is actually choosing among. Alternatives must be distinct (no overlap), actionable (something the decision maker can actually do), and complete (cover all viable options).

- - — *Complete list of alternatives*

Name every alternative, including "status quo." if appropriate. For each: one sentence that will pass the clarity test. .

- - — *Expansion beyond the obvious*

Explain what you did to generate alternatives beyond the first ones that came to mind. Did you interview the decision maker, run a brainstorm, consult outside expertise? Note any new alternative you generated this way.

- *INCLUDE IF APPLICABLE — Options — contingent decisions*

If the decision maker can make follow-on choices after observing an intermediate outcome (e.g., stage-gate investments, real options), describe those here. Show the sequence of decisions in words or as a decision tree stub.

6. Relevant and Reliable Information

This section contains descriptions of information and data sources you referenced.

This section consolidates Tasks 1–3 of HW5 and the probability sub-model from the project draft to explain the full CDN.

- - — *Node list with degrees (clarity test)*

List every node in your Causal Decision Network (CDN). For each node give: (a) its type — uncertainty (circle), decision (square), or value (diamond); (b) its degrees — the mutually exclusive, collectively exhaustive states or range; (c) clarity-test result — state whether a clairvoyant could name the degree with no further interpretation. If you had to refine a node, note what was ambiguous and how you sharpened it.

- - — *Causal Decision Network (CDN) — structure and arcs*

Show your CDN assembled from its three sub-models (probability sub-model, decision sub-model, value sub-model). For every arc, state its meaning in one sentence: an arc into an uncertainty node shows conditioning (relevance); an arc into a decision node is information known at that decision; arcs into the value node show what value depends on. Confirm the network is acyclic and has exactly one final value node with no outgoing arcs.

- - — *Assessed probability distributions*

For each uncertainty node: report its initial P10 - P50 -P90 interval assessment. If you used a tornado diagram, for those nodes that remain, give the assessed distribution over its degrees, conditioned on the degrees of any parent nodes (a conditional probability table (CPT) if the node has parents). State whose judgment the assessment represents (a team member acting for the decision maker, or an outside expert) and the background information ϵ you conditioned on.

- - — *Assessment method and bias controls*

If you used tools to assess probabilities e.g. the probability wheel, explain your choice of method. Name any biases you guarded against (e.g., overconfidence, availability). Do you have a way to check the elicitation calibration?

- *INCLUDE IF APPLICABLE — Value of clairvoyance — observable nodes*

Describe the tests in your model: uncertainty nodes that are observable before a decision is made. Does the test have a cost to balance against it's possible value? Indicate the active path from the test node to the value.

7. Sound Reasoning

This section contains your computational work and your analytic, quantitative results

Assemble the full CDN and discuss how it is solved. This section corresponds to the Sound Reasoning area of Decision Quality and to Task 5 of HW5.

- - — *Roll-back solution*

If you've reduced your model to a decision-probability tree, solve the model by rolling it back. Perform any applications of Bayes rule needed before you create the tree. At each uncertainty node take the expected value; at each decision node take the alternative with the highest expected value. Report each alternative's expected utility.

- - — *Computational methods.*

If you used software, explain how you analyzed your model, what tools or techniques you drew upon.

- - — *Recommended alternatives*

Show the numerical results - e.g. expected utility or cumulative output deals for the alternatives,

- - — *Are there recommendations on additional information collection or discovery results from perfect value of information or test (imperfect) value of information analysis? Which uncertainties does it apply to?*

- *INCLUDE IF APPLICABLE — Are there any options included in the model that would be exercised?*

- *INCLUDE IF APPLICABLE — Dominance*

If any alternative is dominated — yields a worse outcome than another in every scenario — identify it and use dominance to remove it before the roll-back. State whether dominance is deterministic (always worse) or stochastic (worse in expected value regardless of probabilities).

- - *INCLUDE IF APPLICABLE — Sensitivity analysis*

Vary the probability or value assessments to test sensitivity to decision choices. For each, observe how much the parameter must shift before the recommended alternative changes. Name the one or two inputs the decision is most sensitive to, and reflect on whether that sensitivity changes your confidence in the recommendation.

8. Commitment to Action

Briefly tell how the analysis is useful in how it informs the decision. Put yourself in the role of the decision analyst and explain your recommendation to the decision maker. In short make the case for your analysis of how it offers clarity of action to the decision maker.

- - — *Recommended decision*

How would you state the recommendation and any qualifications to the decision maker? Are there contingent decisions, e.g. a “policy” recommendation that arise from multiple decision nodes?

- - — *Are there any assumptions that may need to be investigated that are not included in the analysis? Did the analysis reveal any areas where the model was incomplete?*

- - — *Are there any limitations inherent in this methodology, or ethical concerns you should convey to the decision-maker?*

- *How might the Decision Analysis process itself be refined or improved?*

Appendix

- - — *Credits*

List every published source, or external resource that shaped the analysis: published data, online sources, software, AI tools, and any outside expert you consulted. This is required — omitting it is an academic-integrity issue.

- - — *Programming, code*

Include any scripts, executable notebooks, or spreadsheets you used. For any applications used include the model files generated (e.g. Genie, include your xdsl file).

Submission Checklist

Before submitting, confirm every mandatory item is present. Items marked ○ are required only if applicable to your project (see instructions in each section).

- *[Section 1] Primary decision statement, decision maker, frame (in/out), and purpose*
- *[Section 2] Value measure and units; complete value function or constructed scale*
- *[Section 2] Ethical concerns and tradeoffs identified*
- *[Section 2] Utility elicitation (best/worst, intermediate prospects, certain equivalent) — if applicable*
- *[Section 3] Complete list of alternatives including do-nothing*
- *[Section 3] Explanation of how alternatives were expanded beyond the obvious*
- *[Section 3] Contingent decisions / options — if applicable*
- *[Section 4] Node list with degrees and clarity-test results for every CDN node*
- *[Section 4] CDN diagram with every arc explained; confirmed acyclic with one value node*
- *[Section 4] Assessed probability distributions with source of judgment and conditioning information*
- *[Section 4] Probability wheel method stated; at least one bias named and addressed*
- *[Section 4] Value of clairvoyance computed for at least one observable node — if applicable*
- *[Section 5] Assembled CDN figure*
- *[Section 5] Roll-back solution with recommended alternative and expected value/utility*
- *[Section 5] Sensitivity analysis on key inputs*
- *[Section 5] Dominance identified and used — if applicable*
- *[Section 6] Recommended decision in plain language*
- *[Section 6] Next steps with owners*
- *[Section 6] Key uncertainties to monitor*
- *[Section 7] Summary paragraph (recommendation + key insight)*
- *[Section 7] Credits for all data sources, AI tools, and outside experts*
- *Submit the relevant software outputs like codes, computations, Genie diagram*